

REAL-TIME SYSTEM FOR EMG SIGNAL ANALYSIS OF STATIC AND DYNAMIC CONTRACTIONS

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Abstract - This paper describes a PC/Windows real-time system that uses an advanced data acquisition processor to analyse surface EMG signals. The system is able to calculate time- and frequency domain parameters of surface EMGs during static and repeated isokinetic contractions with a high degree of standardisation. Using a database and Windows/DDE gives flexible opportunity to additional processing. Selection of parameters as well as implementation of future parameters is easy to perform, since the system is software-based.

Keywords: EMG, PC, Muscle fatigue, Static contraction, Dynamic contraction.

INTRODUCTION

Peripheral muscle fatigue and how it influences endurance and/or the risk of injuries and diseases has attracted great interest in the research fields of rehabilitation medicine, ergonomics and sports physiology. It is known from static contractions that the power spectrum of the surface EMG is compressed towards lower frequencies and that the signal amplitude increases during fatigue [1]. The average rectified value (ARV) and the root mean square value (RMS) provides a measure of the number of recruited motor units. The mean frequency (MNF) and the median frequency (MDF) of the power density spectrum will to a great extent be determined by the conduction velocity (CV) of the active muscle-fibre membrane [2].

The aim of the present work was to develop a system for clinical use with the following characteristics:

- **Graphic feedback:** on-line measurement and real-time calculation, including graphic presentation, of surface EMG signals and related biomechanical parameters on a PC/Windows.
- **Flexible parameter selection:** selection of parameters should be easy to perform in the test situation and should be implemented as a software solution.
- **Standardised methods:** besides handling static contractions, the system should be able to analyse repeated isokinetic contractions with a high degree of standardisation with respect to the range of motion.
- **Flexible additional processing:** including handling of patient and measured data using a database, and transferring data to other PC/Windows program (DDE).

IMPLEMENTATION

The data acquisition system (Figure 1) comprises four main parts:

1. A bipolar isolated EMG amplifier for registration of the EMG signals.
2. A dynamometer with analogue signal condition equipment is used as a transducer for the force, position and velocity signals.
3. A data acquisition processor (DAP) is the main processor for sampling and calculation of the data.
4. A personal computer (PC).

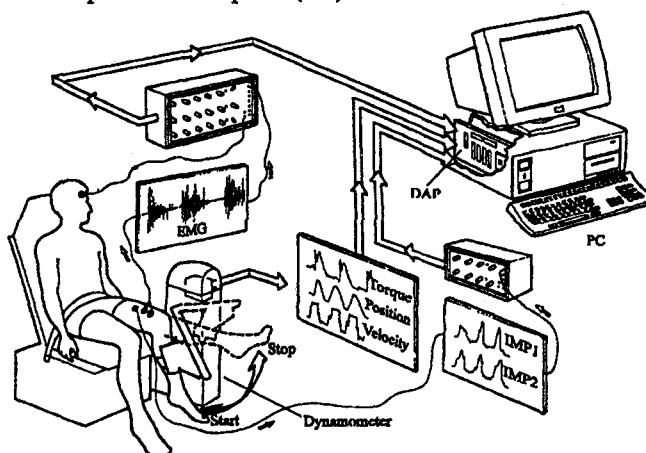


Figure 1: Schematic drawing of the data acquisition system.

Windows is not appropriate for real-time application. Therefore we used a data acquisition processor (DAP) for data acquisition and calculation of all parameters. The processed data were buffered before being sent to the PC, which freed the PC to perform other functions momentarily, e.g. displaying and saving data and processing commands from the operator. All signals were amplified and analogue-to-digital converted with 12-bits accuracy in the signal range $\pm 5V$. The sampling rate can be selected individually. The usable energy of the EMG signal is limited to the 0 to 500 Hz frequency range, with the dominant energy in the 50-150 Hz range. The EMG signal should therefore be sampled with at least 1 kHz. The power-density spectrum was obtained using the FFT technique. The sampling rate, pre-windowing, FFT block length, number of overlapping windows and percent of overlapping, can easily be changed to increase spectral resolution and to reduce the variance.

All processed parameters are displayed in real-time in user selectable number of graphs. Data (patient, parameters

and raw-data) could be stored in a database for later displaying, printing or additional processing (Figure 2).

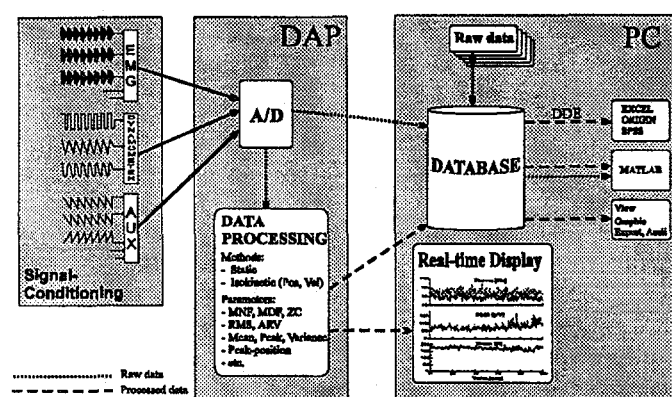


Figure 2: Block schema of software and hardware.

During a static contraction signals are divided in time segments and the selected parameters were calculated for every segment during the test. The system is also able to handle isokinetic contractions. Usually, maximal (or at least near maximal) contractions are investigated. Besides EMG parameters, we also calculated peak-torque, work and peak torque position for each contraction cycle. One contraction cycle is defined as the total movement throughout the predetermined range of motion and back again. A contraction cycle has an active (contraction) and a passive (relaxation) phase or two active phases.

If the time and frequency parameters are displayed on the computer screen in real-time, they will express data of a random nature, since they describe biological data with great variance. Consequently, these data curves are difficult to trace easily on the screen with the eye. To overcome this problem, we chose smoothing to suppress high-frequency fluctuations. The exponentially weighted moving average (EWMA), $\hat{y}(t+1) = \lambda y(t) + (1-\lambda)\hat{y}(t)$, will give less and less weight to data as they get "older" and "older". Depending on the value of λ , the EWMA will give more or less weight to older data ($\lambda=1$: no weight to older data). A value of 0.3 for the constant λ has proved to give a good estimate [3].

RESULT AND DISCUSSION

A system based on PC/Windows that is capable of on-line and real-time acquisition and analysis of EMG signals and other common biomechanical parameters for different types of muscle contractions (static and isokinetic) has been developed. The system has been validated for both known and physiological signals (i.e. surface EMG-signals) and the results were as expected [4].

To produce a time-frequency representation of a signal, the short-time Fourier transform can be used. A major drawback of this method is that a stationary signal is assumed. Even when there is no voluntary change of muscle

state, myoelectric signals are non stationary simply due to the inherent physiology of the organ. It is a common understanding that during a static contraction the surface myoelectric signal is locally stationary. During isokinetic contraction the frequency parameters can be questioned. However, even though stationarity could not be assumed, this is a relatively common clinical way of indicating fatigue from an EMG point of view.

The present software solution will provide an opportunity to select other methods of calculating parameters in the future. A relatively new technique, the wavelet transform, is a technique that does not assume that the signal is stationary, and could be an interesting method for isokinetic conditions.

With the presented system it is possible to avoid the part of the contraction associated with motor unit recruitment and the acceleration/deceleration parts of the contraction. The system will allow well-defined registration of relations between the EMG and mechanical parameters during dynamic contraction.

A good situation exists for adequate graphic feedback of the investigated subject throughout the test. The system can also be used in such a way that the subject follows his/her performance continuously on the computer screen. It will be possible to monitor any combinations of the calculated parameters in real-time on the screen of the PC. Merletti et al. have proposed a new regression-free index of muscle fatigue as ratio of the area above and below the measured data points [5]. This index is very suitable to this real-time system because new values can be updated without recalculation.

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